6710503739_HW#7_1

August 30, 2026

1 HW7 - Question 1

[28]:
```
import pandas as pd
from sklearn.neighbors import KNeighborsClassifier

data = [
    ("Red",    "Sport", "Domestic", "Yes"),
    ("Red",    "Sport", "Domestic", "No"),
    ("Red",    "Sport", "Domestic", "Yes"),
    ("Yellow", "Sport", "Domestic", "No"),
    ("Yellow", "Sport", "Imported", "Yes"),
    ("Yellow", "SUV",   "Imported", "No"),
    ("Yellow", "SUV",   "Imported", "Yes"),
    ("Yellow", "SUV",   "Domestic", "No"),
    ("Red",    "SUV",   "Imported", "No"),
    ("Red",    "Sport", "Imported", "Yes"),
]
df = pd.DataFrame(data, columns=["Colour", "Type", "Origin", "Stolen"])
df
```

[28]:

	Colour	Type	Origin	Stolen
0	Red	Sport	Domestic	Yes
1	Red	Sport	Domestic	No
2	Red	Sport	Domestic	Yes
3	Yellow	Sport	Domestic	No
4	Yellow	Sport	Imported	Yes
5	Yellow	SUV	Imported	No
6	Yellow	SUV	Imported	Yes
7	Yellow	SUV	Domestic	No
8	Red	SUV	Imported	No
9	Red	Sport	Imported	Yes

1.1 1. Encode the attributes

[29]:
```
X = pd.get_dummies(df[["Colour", "Type", "Origin"]]).astype(int)
y = df["Stolen"]

query = pd.DataFrame([{"Colour": "Red", "Type": "SUV", "Origin": "Domestic"}])
Xq = pd.get_dummies(query).reindex(columns=X.columns, fill_value=0).astype(int)
```

```
X.head()
```

[29]:

	Colour_Red	Colour_Yellow	Type_SUV	Type_Sport	Origin_Domestic	Origin_Imported
0	1	0	0	1	1	0
1	1	0	0	1	1	0
2	1	0	0	1	1	0
3	0	1	0	1	1	0
4	0	1	0	1	0	1

1.2 2. Train KNN and predict

k = 5

[30]:
```
knn = KNeighborsClassifier(n_neighbors=5, metric="euclidean")
knn.fit(X, y)

pred = knn.predict(Xq)[0]
proba = knn.predict_proba(Xq)[0]

print("Prediction for (Red, SUV, Domestic):", pred)
```

```
Prediction for (Red, SUV, Domestic): No
```

1.3 3. Which neighbours produced that vote?

[31]:
```
dist, idx = knn.kneighbors(Xq, n_neighbors=5)
neighbours = df.iloc[idx[0]].copy()
neighbours["Distance"] = dist[0].round(3)
neighbours
```

[31]:

	Colour	Type	Origin	Stolen	Distance
0	Red	Sport	Domestic	Yes	1.414
1	Red	Sport	Domestic	No	1.414
2	Red	Sport	Domestic	Yes	1.414
7	Yellow	SUV	Domestic	No	1.414
8	Red	SUV	Imported	No	1.414

1.4 Answer

Their majority label is No, so the KNN prediction for (Red, SUV, Domestic) is Stolen = No.

the $k = 5$ result is the one that actually reflects the data.